

Evaluating Generative Agent Simulation Performance

Business question: Can a Human simulation be trusted to represent the specific real person it is intended to mimic?

The case concerns a Beauty & Wellbeing company that created LLM-based Human simulations from interviews with young business professionals. The evaluation therefore focuses on **individual-level behavioral fidelity**, not generic answer quality: does the simulation preserve what that person actually says, prefers, does and values when answering held-out questions?

1. Why this matters?

Human simulations can generate plausible consumer responses at scale, but plausibility is not the same as fidelity. A simulation may stay on-topic while omitting an important motivation, reversing a preference or introducing unsupported personal detail.

The key question is therefore: **Does the simulated response reproduce the individual, rather than merely resemble a plausible consumer?**

The supplied test set provides a direct held-out evaluation: **30 Human-AI response pairs across 3 individuals, with 10 open-ended questions per person.**

2. Evaluation Concept

Fidelity dimension	What is tested	Why it matters
Core	Central answer / meaning	Tests whether the essential response is preserved
Behavior	Actions, habits, and context	Tests whether the simulation preserves what the individual actually does
Preference	Choices and priorities	Tests whether individual preferences and decision patterns are preserved
Motivation	Reasons behind choices	Tests whether person-specific reasoning is retained rather than replaced by generic plausibility
Nuance	Qualifiers, trade-offs, and uncertainty	Tests whether subtle individual context is preserved rather than flattened into a generic response

G-Eval Core Fidelity is the primary automated 1–5 score. The remaining dimensions provide diagnostic evidence about why a simulation succeeds or fails.

3. Method

- **Measure.** G-Eval provides the primary 1–5 behavioral-fidelity score, while BERTScore adds a contextual semantic-similarity diagnostic.
- **Explain.** Claim extraction and alignment identify aligned, partial, contradicted, missing, and unsupported information; NLI independently cross-checks contradictions.
- **Validate.** All 30 pairs are manually reviewed using a fixed 1–5 rubric with contradiction, important-omission, and unsupported-detail flags.
- **Synthesize.** Automated evaluation, Human review, automated-vs-Human validation, and person-level consistency are combined in the final assessment without creating a weighted composite score.

4. Evaluation Principles

- **Pairwise first:** Every AI response is evaluated against its corresponding Human response before any aggregation.
- **G-Eval:** It is the primary automated score; the remaining signals provide complementary diagnostic evidence.
- **BERTScore:** It measures contextual semantic similarity, not behavioral correctness.
- **NLI:** It is an independent contradiction cross-check, not the primary fidelity evaluator.
- **Unsupported means unsupported by the Human reference;** it does not by itself mean false or hallucinated.
- **Important omissions, qualifiers, and uncertainty are assessed explicitly** rather than treated as simple similarity differences.
- **Automated and Human judgments remain separate,** so disagreement is visible and can be examined rather than averaged away.

5. Key Findings

Decision question	Evidence	Result / implication
How faithful are the simulations overall?	Automated and Human-reviewed match scores	2.63 / 5 automated; 2.53 / 5 Human-reviewed; partial overall match
Where do they break?	Human-reviewed failure modes	73.3% important omission; 36.7% contradiction; 100% unsupported detail
Is person-specific information preserved?	Claim-level analysis	Of 121 Human claims, 15 aligned, 35 partial, 10 contradicted, and 61 missing
How much unsupported information is introduced?	AI claim analysis	198 of 254 AI claims are unsupported by the corresponding Human reference
How well does automated evaluation align with Human review?	Automated-vs-Human validation	50% exact agreement; 100% within one point; $\rho \approx 0.82$
How well does contradiction detection align with Human review?	Claim-based and NLI checks	Both achieve 90% row-level agreement

The simulations frequently preserve the broad topic or direction of an answer, but person-specific fidelity remains limited.

6. What the failures reveal?

Three representative failure patterns illustrate the main weaknesses:

- **Behavioral reversal:** the Human says, *“I don’t usually buy in bulk...”* while the simulation says, *“I regularly buy toothpaste and deodorant in bulk...”*
- **Correct choice, unsupported reasoning:** the simulation reaches the same choice *“option 2”*, but adds reasons that were never stated in the Human reference.
- **Preference mismatch:** the Human names Keira Knightley if forced; the simulation selects a different celebrity and adds supporting rationale.

These cases show that semantic similarity alone is insufficient, a response can remain on-topic, or even preserve the final choice, while still misrepresenting the individual.

Decision implication: The simulations preserve broad directional themes, but should not be treated as faithful substitutes for specific individuals without validation.

Recommended model: automated screening → flag contradictions, omissions, and unsupported detail → Human review for high-impact or ambiguous cases.

Potential improvements: stronger person-specific grounding, explicit uncertainty handling, larger validation samples, multiple reviewers, and further evaluator calibration.

7. Implementation, Reproducibility and Limitations

The evaluation is implemented as a modular Python pipeline with configurable models, structured Pydantic outputs, cached model calls, explicit runner scripts, reproducible intermediate outputs, and automated tests. Final outputs retain row-level automated evidence, Human-review results, and aggregate summaries, allowing each conclusion to be traced back to its underlying evidence.

The supplied dataset contains only **3 people × 10 questions**, so the findings should not be generalized beyond this subset. G-Eval can be sensitive to model and prompt design; NLI can be sensitive to scope and context; BERTScore measures semantic similarity rather than behavioral correctness; and manual review remains subjective despite the use of a fixed rubric.

8. Implications

The main challenge is not making a simulation sound human, it is making it remain faithful to the specific Human. Across the held-out test set, the simulations often preserve broad themes but frequently omit person-specific information, introduce unsupported detail, and in some cases reverse stated behaviors or preferences.

The evaluation framework provides a measurable **trust layer** around Human simulations: it quantifies fidelity, explains where simulations deviate, validates automated judgments against Human review, and identifies where automated assessment is sufficient versus where Human validation remains necessary.